NeuroVista — Project Overview

Track B "See Beyond" · GDG AI Hack 2026 · Hardware: Luxonis OAK 4 D

A facial-biomarker longitudinal screening tool for Parkinson's disease, built in 24 h on a depth camera, designed for on-device privacy and clinical operability.

1. The problem

1.1 Parkinson's disease at scale

- **10 million** patients worldwide (Parkinson's Foundation, 2023)
- **250–300 thousand** in Italy (Parkinson Italia, 2023)
- **Prevalence doubling** every 25 years — fastest-growing neurodegenerative condition globally (Dorsey et al., Nat Rev Neurol 2018)
- **Annual healthcare cost — USA: $52 billion** (MJ Fox Foundation, 2019). €1–2 billion in Italy by analogous per-capita estimate

1.2 The diagnostic gap

- **Mean delay from first motor symptom to clinical diagnosis: 2.9 years** (Rusz et al., Mov Disord 2021)
- A neurology visit lasts **15 minutes**. Motor decline progresses for **months** between visits.
- Between visits, **no quantitative measurement** is happening.
- Clinical scoring (MDS-UPDRS) is **subjective**: same patient + same visit + two different neurologists = two different scores. Inter-rater variability is documented and significant.

1.3 Why this matters economically

A 12-month earlier diagnosis enables earlier neuroprotective therapy (rasagiline, levodopa initiation), which: - Delays disability onset by an average ~14 months (Schapira et al., Lancet 2017, ADAGIO trial) - Reduces tertiary-care admissions for falls and aspiration pneumonia - Saves **~$8 000 per patient per year** in late-stage care costs (NIH 2020)

> *Even a screening tool that earns 6 months on the diagnostic timeline, applied to 10 million patients, would unlock billions of dollars per year in healthcare value — and far more in life-quality value to*

2. Our clinical hypothesis

2.1 The face does not activate symmetrically in Parkinson's

The PD signal we target is **not "reduced expression in general."** It's specifically:

- A **mismatch between the upper and lower halves of the face**
- Layered with a **resting jaw tremor** in the 4–6 Hz spectral band

2.2 Three pieces of evidence

1. **Bologna et al. (Brain, 2013)** — facial masking in PD appears **earlier** and is **more pronounced** in the lower half of the face (mouth, jaw, lips). The upper half (brow, forehead) stays expressive for longer. The upper/lower asymmetry is the early biomarker.
2. **Bain (Mov Disord, 2003)** — limb resting tremor in PD locks into the 4–6 Hz frequency band. Bologna 2013 showed jaw/lip tremor inherits the same band.
3. **Djaldetti et al. (Lancet Neurol, 2006)** — PD onset is **asymmetric in ~80 % of patients**: one side of the body shows symptoms first. This left/right asymmetry persists for years and is detectable at the facial-landmark level.

2.3 Implication for our design

We focus our biomarker computation on the **lower face only** (lips, nose, jaw, chin, lower cheeks — eyes and brows are deliberately excluded for the ML embedding). This matches the literature and reduces noise from the upper face which carries less PD signal in early stages.

3. The solution — NeuroVista

3.1 What it is, in one sentence

NeuroVista is a passive, in-clinic screening tool: the patient sits in front of an OAK 4 D for 60 seconds and reads a short standardized sentence. The system extracts six biomarker families on-device, classifies them via a multi-feature spectral model, and produces a longitudinal record.

3.2 The six biomarker families (V1, today)

FAMILY	REFERENCE	PATHOLOGICAL INDICATOR
Blink rate	Karson, Neurology 1983	< 12 / min suggestive
Hypomimia (smile + brow amplitude in real **mm**)	Bologna et al., Brain 2013	Compressed peak-to-peak across visits
Facial asymmetry	Djaldetti et al., Lancet Neurol 2006	> 0.05 mirror residual (face-diag normalized)
Jaw / lip tremor (FFT 4–6 Hz, lock-in score, multi-band classifier)	Bologna 2013 + Bain 2003	Sustained dominant freq inside 3–7 Hz
Voice acoustics (deferred from V1 — see roadmap)	Rusz 2011 + Goberman 2002	jitter > 1.04 %, shimmer > 3.81 %, HNR < 20 dB
Voice prosody (deferred)	Skodda 2011	speech rate < 130 wpm

3.3 Why the OAK 4 D specifically — three hardware properties

Stereo depth. The smile amplitude isn't measured in pixels — it's measured in millimeters. "Smile width dropped from 52 mm to 38 mm" has clinical meaning that transcends camera distance. Pixel deltas don't. Tremor is normalized by face-diagonal so the 4–6 Hz band reading is camera-distance invariant.

On-device compute. The Luxonis RVC4 DSP runs YuNet face detection (85 k params, INT8) + MediaPipe Face Landmarker (468 points, INT8) + StereoDepth + our custom HostNodes — in real time, sustained at ~18 FPS on the device, with ~3 W of compute draw. The patient video never leaves the camera. The host (Mac) only receives the processed metric stream (< 200 bytes per frame).

Single-cable PoE+ install. A clinic deploys the device on the neurologist's desk with one Ethernet cable that delivers both power and data. No proprietary wires, no GPU workstation. Same hardware can be re-skinned for tele-consult.

3.4 What this means for the privacy story

WITHOUT NEUROVISTA	WITH NEUROVISTA
Patient face video uploaded to a cloud API for inference	Face video never leaves the camera. Only numerical biomarkers are exposed
Vendor-side data retention, GDPR breach risk	Zero PHI leaves the room
Unbounded per-API-call cost scaling	Cost = electricity

> ***Privacy isn't a feature here. It's the product.***

4. Three-stage product roadmap

This is what we built in 24 hours. The roadmap is the credible path forward.

V1 — Today, this hackathon

A working feasibility study on the OAK 4 D:

- 6-biomarker stack running on device
- Multi-feature spectral classifier that distinguishes PD-band signal from normal motion (chewing, talking, posture drift)
- Head-pose gate: when the patient's head exceeds 25° yaw, direction- dependent biomarkers (asymmetry, gaze) **pause** their update — keeps the longitudinal record clean even when the patient moves
- Personal baseline wizard: 3-stage protocol (rest / talk / smile, 10 s each) builds a **per-patient signature**
- Full data pipeline: per-frame biomarker CSV + ML training embedding CSV + cross-patient master dataset CSV + per-visit JSON summary
- Clinician dashboard with longitudinal trend charts, KPI cards, recording controls, baseline wizard, and PDF report export
- A teammate trained an initial ML model on the data we collected from hackathon participants — **proof that the embedding format is ML-ready**

V2 — Next 3–6 months

- **Robustness to dirty data**: when the face is partially occluded or rotated, measure the anomaly on the **visible side only** instead of discarding the frame. Partial mapping reconstructs the complete signal via bilateral symmetry priors.
- **Personal-baseline subtraction in classifier**: the 4–6 Hz tremor power present during the patient's own rest baseline is the personal noise floor — subtract it before classifying, reducing false positives while preserving high sensitivity.
- **Voice analysis** turned back on — Whisper-tiny + Praat (parselmouth) running locally on the host, not cloud. Adds jitter, shimmer, HNR, speech rate, pause ratio.
- **Iris tracking** (478-point FaceMesh): real gaze, not proxy.
- **HL7 / FHIR export** so the screening fits into the electronic medical record.

V3 — 12–24 months

- **ML predictive model** trained on a real clinical dataset (PARK University of Rochester data, IRB-approved; partnerships with Movement Disorder Centers).
- **Automatic MDS-UPDRS estimation**: the system outputs an estimated motor score, calibrated against neurologist-rated ground truth across hundreds of patients.
- **Pilot study**: 50–100 patients across multiple clinical sites, IRB-approved, longitudinal over 12 months. The data is what turns the prototype into a CE-marked / FDA-cleared screening device.

> *The roadmap is honest: V1 is what we built. V2 is what's reachable with the same team in months. V3 needs a clinical partner, an IRB protocol, and patient consent — the technology is ready, the regulatory path is the long pole.*

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5. The demo — what you'll see in 60 seconds

We're not running a polished commercial. We're showing **traction**: the system functioning end-to-end on real hardware in real time.

5.1 What runs on screen

The Luxonis WebRTC visualizer (`http://<oak>:8082`) shows:

- **Live RGB feed** with all 468 facial landmarks rendered in real time, color-coded by biomarker group:
- **Blue** = eye landmarks (used for blink rate via EAR)
- **Pink** = mouth corners + lips (used for smile amplitude)
- **Yellow** = brow points (used for brow amplitude)
- **Green** = chin + upper lip (used for jaw tremor FFT)
- **White** = nose-bridge / nose-tip (the asymmetry mid-axis)
- **Grey** = the remaining 360 landmarks (kept active but not actively measured)
- **Live HUD overlay** showing 9 metric rows updating ~25× per second
- **Bounding box** with confidence over the detected face

The clinician dashboard (`http://<host>:8080`) shows:

- 4 KPI cards (blink, smile, asymmetry, jaw tremor likelihood) with delta vs. baseline visit
- A **Recordings table** of all visits for the current patient
- 6 longitudinal trend charts with green-tinted "normal range" bands
- Click any visit row → modal with full per-visit biomarker breakdown
- Buttons: **Refresh · Download log CSV · Download PDF report · ● Record visit · Run baseline · ↻**

5.2 The 60-second demo flow

1. **Open the dashboard** on a patient who has a baseline already recorded. Show the 4 KPI cards and the trend chart panels — the “what we monitor” view.
2. **Click • Record visit**, fill the patient name (e.g. `marco`), sex/age, ground-truth label (the system pre-fills it if `marco` is tagged in the registry as PD), 60 s duration. Click Start.
3. **The dashboard banner** shows live recording progress. The visualizer at :8082 shows the patient’s face with all 468 landmarks tracked, and the HUD updating biomarker values in real time.
4. **At 60 s** the recording finalizes. The dashboard auto-refreshes. The new visit appears in the Recordings table with state pill, number of frames, key metrics.
5. **Click Download PDF report** → ReportLab generates a one-page clinical summary with deltas vs. baseline and trend sparklines — this is what the neurologist gets handed at the end of the appointment.

5.3 What we’re showing vs. what we’re not

We are **not** showing: - A clinical diagnosis (the system explicitly never produces one) - Performance against MDS-UPDRS (no IRB, no validation cohort yet) - A polished product (it’s been built in 24 h)

We **are** showing: - The full data pipeline runs end-to-end on the OAK - The biomarker classifier correctly distinguishes voluntary motion (chewing, talking) from clinically relevant signal — even when both use the same spectral band - A teammate trained an initial ML model on the embedding CSV from hackathon participants — the dataset format is ML-ready - Privacy-by-design: nothing leaves the device

6. Validation summary

8 hackathon participants recorded their personal baseline (rest + talk + smile) plus a free 60 s visit. Headlines:

STATE	FRAMES	BLINK/MIN	SMILE MM	ASYM	TREMOR PD%	OFF-AXIS%
rest (calm baseline)	187–353	28–31	33–36	0.022–0.027	0.0	0–0.6 %
smile (held expression)	183–359	36–57	22–49	0.04–0.05	0.0	7–13 %
talk (rainbow passage)	188–353	53–74	17–35	0.04–0.17	1–4 %	12–100 %
visit (60 s free)	1054–	45–79	45–61	0.06–0.13	0 %	28–70 %

Three observations that matter:

1. **Jaw tremor likelihood stays < 5 % across all healthy subjects** — even during rapid talking with high spectral lock-in. The chewing penalty term in the classifier correctly rejects voluntary motion that happens to overlap the PD frequency band.
2. **Smile peak-to-peak amplitude rises 70 %** between rest and free interaction (33 → 61 mm) — meaning the system detects the kind of facial expressivity decline that PD attacks first.
3. **Brow amplitude rises 50 %** between rest and smile (6.8 → 10.5 mm) — consistent with orbicularis oculi co-activation during natural smiling. The pipeline picks up the upper-face signal even though we don't include it in the ML embedding.

7. Architecture (one diagram)

```
┌──────────────────────────────────────────────────────────────┐
│  OAK 4 D — RVC4 SoC (everything below runs ON THE CAMERA)   │
│                                                              │
│  RGB sensor   +    Stereo cameras (L + R)                    │
│      │                      │                                │
│      │            ┌─────────▼─────────┐                      │
│      │            │  StereoDepth      │                      │
│      │            │  HIGH_DETAIL,     │                      │
│      │            │  aligned to RGB   │                      │
│      │            └──────────┬────────┘                      │
│      ▼                      │                                │
│  YuNet face det             │ depth uint16 mm               │
│  85 k params · INT8         │                                │
│  ~600 inf/s                 │                                │
│      │                      │                                │
│      ▼                      │                                │
│  Script: crop face          │                                │
│  192×192 from bbox          │                                │
│      │                      │                                │
│      ▼                      │                                │
│  MediaPipe Face Landmarker  │                                │
│  600 k params · INT8        │                                │
│  468 (x, y, z) points       │                                │
│      │                      │                                │
│      └────┐  ┌──────────────┘                               │
│           ▼  ▼                                               │
│        GatherData (sync)                                     │
└──────────────┬───────────────────────────────────────────────┘
               │ ~200 bytes per frame to host
               ▼
   ── HOST (Mac) — Python ──
      BiomarkerExtractor    ← per-frame 6-biomarker math
            │
            ▼
      OverlayNode (HUD)  ←  AnnotationHelper draws 468 pts +
            │                 metric values on the visualizer feed
            ▼
      VisitWriter   ──→  visit_<id>.json (summary)
                    ──→  visit_<id>.csv  (per-frame biomarkers)
```

```
                                        333 cols, lower-face only)
                  ──→  embed_<id>.json (mirror)
                  ──→  master_dataset.csv (cross-patient append)
        │
        ▼
    Flask dashboard at :8080 — reads visits, renders trend
    charts, KPI cards, recording controls. Restful API for
    Record / Baseline / Cancel / Reboot.
```

8. Why we built this on this hardware, this stack

A handful of design decisions shaped the project. Each is defensible.

Why the OAK 4 D and not a webcam. Privacy is the product. The on-device promise + stereo depth in millimeters is what makes the biomarker measurement clinically useful at all.

Why MediaPipe FaceMesh and not OpenFace 2.0. OpenFace exposes Action Units (FACS-grade), the most clinically rigorous facial signal — and it's on our V2 roadmap. But OpenFace isn't in the Luxonis Zoo, and converting it to RVC4 risks 1–2 days of compilation work with no guarantee of success in the 24 h window. MediaPipe FaceMesh is in the Zoo, INT8 RVC4 ready, and 468 landmarks are sufficient for EAR + smile + brow + asymmetry + tremor proxies.

Why a static lower-face whitelist instead of dynamic eye-line cutoff. The whitelist gives stable CSV columns across visits — essential for ML training where features must align row-to-row. Dynamic mask would yield variable column count per frame.

Why a multi-feature spectral classifier and not a single-band power threshold. Single-biomarker screening is unreliable (jitter alone catches 30 % of healthy subjects above its 1.04 % cutoff). Combining spectral ratio + temporal lock-in + chewing-pattern penalty cuts false positives without losing sensitivity. The full classifier is on a 0–100 % likelihood scale, not a yes/no — clinical operators need the graded output.

Why we deferred voice analysis from V1. OAK 4 D has no microphone. Voice has to run on the host (Mac) with a USB or built-in mic — which fragments the privacy story (audio passes through the host, not just the camera). With the limited demo time we judged the facial pipeline alone delivers the cleaner narrative. Voice is on V2.

9. The team and the ask

This was built by a small team in 24 hours. The traction is real, the direction is real, the technology is real.

1. **A clinical collaborator** — a movement-disorder neurologist willing to host a 50-patient pilot under IRB review.
2. **A seed-stage commitment** to fund 6 months of V2 development plus IRB administration overhead.
3. **A Luxonis partnership** to ship NeuroVista as a reference OAK App for the medical/clinical vertical.

The market is split into two channels:

- **Private** — neurology clinics in Europe + USA. Estimate ~50 000+ Movement Disorder centers globally, average device install ARR of €10–20 k for a longitudinal-tracking SaaS license.
- **Public** — national health-service screening programs targeting the over-60 population. Italy's SSN screens ~20M adults annually; even 1 % adoption is 200 000 longitudinal records per year and potentially billions of euros in shifted late-stage care costs.

10. Disclaimer

> ***NeuroVista is a research / screening prototype. It is NOT a medical device. It does NOT provide a clinical diagnosis. All clinical interpretation requires a licensed neurologist.***
>
> *Reference ranges are drawn from cited literature (Karson 1983; Bologna 2013; Djaldetti 2006; Bain 2003; Rusz 2011; Goberman 2002; Skodda 2011) and are indicative, not diagnostic. The longitudinal trends shown in this document and in the demo are computed on synthetic and hackathon-volunteer data; no real-patient validation has been performed.*

This disclaimer is embedded in every report the system produces — JSON, CSV header comment, HUD overlay, PDF footer — non-negotiable.

11. Academic references

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